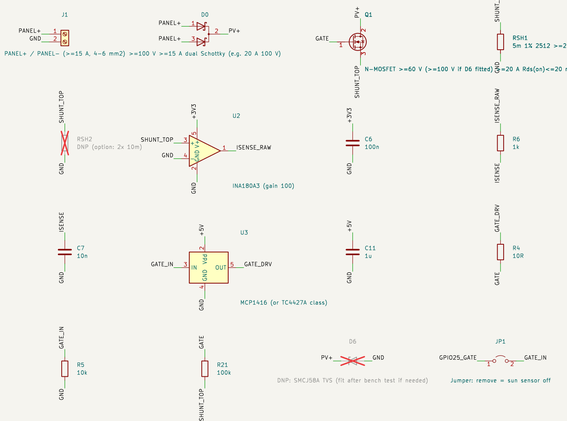
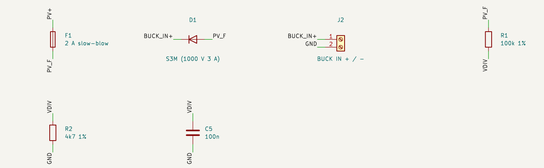
PVA carrier board v1 — schematic draft generated from hardware/CARRIER.md. Label style: nets join by name. Engineer review required before manufacture.

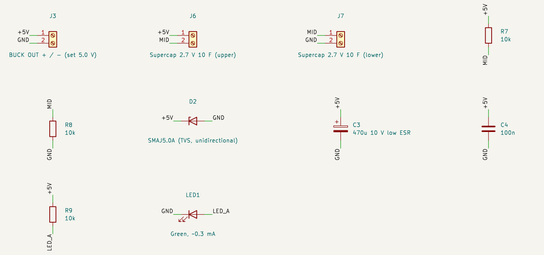
1. Panel input and sun sensor



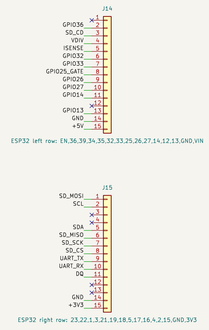
2. Fuse, diode, divider



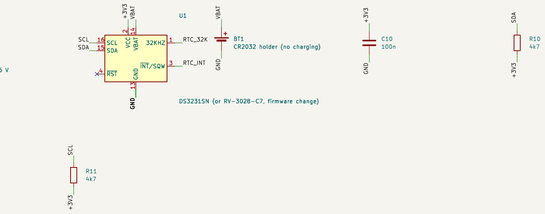
3. 5 V rail



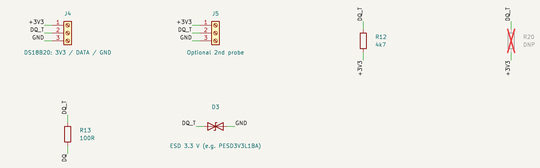
4. ESP32 socket (ELEGOO 30-pin, plugged in; antenna keep-out)



5. Clock



6. Temperature probes

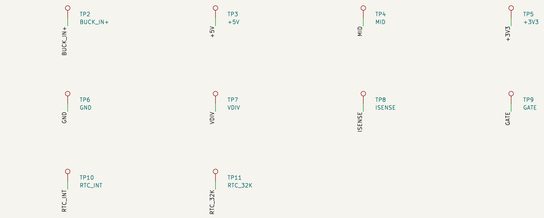


7. microSD (future-proofing)

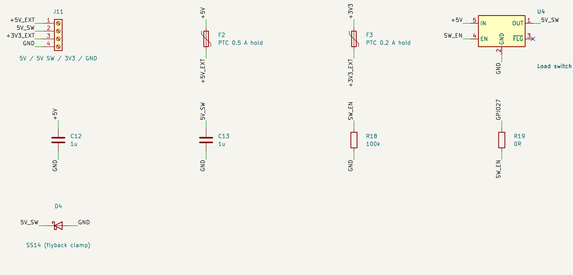


8. Expansion headers and test points





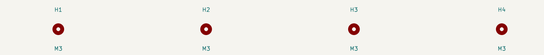
9. Power take-off for add-ons



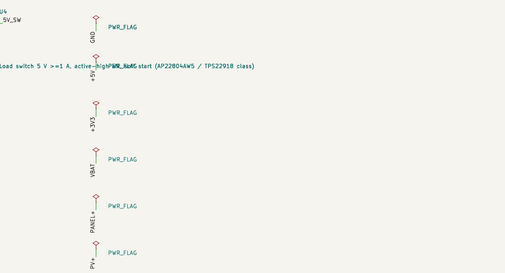
10. AUX power out



Mechanical



Power flags (ERC)



Current paths

1. SUN-SENSOR LOOP: J1 -> D0 -> Q1 -> RSH1 -> GND
   Panel short-circuit current, ~1 ms every 10 s. Panel Isc 5.40 A (label), max ~6 A in cold strong sun.
   A vertical panel peaks near 1.8 A at noon. The panel cannot supply more than its Isc.
2. BOARD SUPPLY: PV+ -> F1 (2 A) -> D1 (3 A) -> buck -> +5V
   Under 0.3 A continuous. F1 and D1 are NOT in the measured current path;
   F1 sits after the Q1 node so the short-circuit pulses never pass through it.
3. AUX (optional add-on): PV+ -> F4 (fitted only with an add-on: <= 3 A on v1. 9 A only with D5 on a heatsink) -> D5 -> J12
   Never through the shunt. D0 and D5 need cooling above ~2 A.

J1 and J12 `>=` 15 A` is the terminal and wire rating, not a measured current.

Measurement ranges

ISENSE (GPIO 35): RSH1 5 mOhm x INA180A3 gain 100 = 0.5 V/A.
   Measured range 0–6 A (pulsed). 1.8 A -> 0.9 V; 5.4 A -> 2.7 V. Output clamps at the 3.3 V supply (~6.6 A): harmless.
   Gain 100 is deliberate: low vertical-panel currents stay well inside the ADC's range.
   Shunt dissipation 0.18 W only during each 1 ms pulse; 2 W rating is for pulse robustness.
VDIV (GPIO 34): R1 100k / R2 447 = × 223 (R2 was 546, changed after review).
   Panel Voc ~44.5 V -> 2.00 V: cold morning 48 V -> 2.15 V: board rating 60 V -> 2.69 V.
   ESP32 ADC accurate to ~2.45 V (= 54 V panel): pin absolute max 3.6 V. R1 limits any fault current to < 1 mA.
   Firmware: DIVIDER = (100 + 4.7) / 4.7 (pva-fw-6.1.1).

ESP32 module and Q1 gate (review 2)

The ESP32 is a complete ELEGOO dev board plugged into J14/J15. Its EN pull-up, reset and boot buttons, USB-serial converter and strapping resistors are on that board, not on the carrier.
EN and GPIO 0, 1, 2, 3, 12, 15 are deliberately not brought out. GPIO 5 (strapping) is held high by R14.
Q1 gate: U3 drives it when +5V exists; R21 (100k, gate→source) holds Q1 off before that, at dawn.
D6 (TVS on PV+): footprint only; fitted if the v1 bench test shows turn-off overshoot. Then Q1 >= 100 V.

Draft: engineer review required before manufacture
Generated from hardware/CARRIER.md by gen_schematic.py
photovoltaic-art / SUNDIAL
Sheet: /
File: carrier.kicad_sch

Title: PVA carrier board v1

Size: A1	Date: 2026-10-20	Rev: draft-1
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1